

Polar regions

The Arctic and Antarctic are the coldest places on earth. The Arctic is a partly frozen ocean, hemmed in by large expanses of windswept tundra; the Antarctic is an ice-covered continent, surrounded by the world's stormiest seas. They are similar to each other—and unlike any other habitat—in that they have 24-hour daylight in summer and perpetual darkness in winter, but they are physically different in ways that have important effects on animal life. In the Arctic, many animals live on land; in the Antarctic, animal life is based almost entirely in the ocean.

Arctic and tundra

Covering about 4.6 million square miles (12 million square km), the Arctic Ocean is both the smallest and the shallowest ocean in the world. For several months in summer, permanent daylight produces a constant supply of energy, which is harnessed by vast quantities of planktonic algae. These form the first link in the Arctic Ocean's food chain, which ultimately nourishes animals as large as whales and polar bears.

Sea ice—or the lack of it—is a major factor in determining where large mammals live, especially during winter when the surface area of the ice is at its greatest. Polar bears and Arctic foxes can

traverse the ice to find food, but seals and some other marine mammals must maintain breathing holes to survive.

Despite the icy conditions, sea life is plentiful in the Arctic because cold water is rich in oxygen and the seabed sediment is rich in nutrients. On land, though, intense winter cold means that

trees cannot survive. The result is tundra—an open, often featureless, landscape, scraped smooth by glaciers during the last ice age. Today, Arctic glaciers are restricted mainly to mountains and to the ice cap that covers Greenland, but large areas of tundra remain permanently frozen underground. This frozen zone—the permafrost layer—prevents spring melt water from draining away, creating waterlogged landscapes in a region where rainfall, or snow, is paradoxically very low.

In late spring and early summer, tundra plants grow, and flower, very rapidly. Geese and other migratory birds arrive to breed, and vast numbers of mosquitoes emerge from tundra pools. The migrants' departure, when the short summer draws to a close, marks the end of another biological year.



SNOWY OWLS are forced to hunt by daylight during the weeks of high summer.



WILLOW GROUSE, like many other animals, turn white in winter to blend in with the snow.



MUSK OXEN and other tundra grazers reach food by scraping away snow with their hooves.



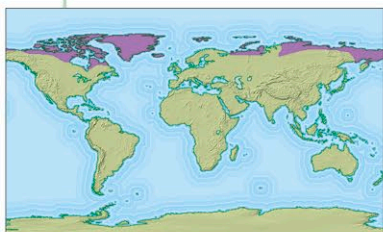
LEMMINGS provide food for many of the Arctic's predators, including foxes.



MOSQUITOES, which emerge in early summer, make life uncomfortable for mammals and birds.



THE POLAR BEAR is a superb swimmer, and it is as much at home on shifting pack ice as it is on tundra.



TRUE TUNDRA is found north of the Arctic Circle, but tundra-like conditions exist on some mountains elsewhere.

LAND

FRESH WATER

